Group 10 - Project: Cloud Infrastructure Migration

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CS 4843 - Cloud Computing

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May 2, 2025

Introduction

Migrating your infrastructure to the cloud is no small feat, there are many factors you must consider and a plethora of options regarding this process. This document covers the approach our team would best fit the needs of the client. We selected AWS for the cloud platform due to its scalability, security, and cost efficiency. The migration began with handling the core networking services by transitioning to AWS Direct Connect. After this, we handled business-critical systems such as moving email to Amazon's WorkMail. Databases were migrated to Amazon Relational Database Service (RDS), and EC2 is utilized for application servers. Lastly, we also decided it would be best to implement EBS for volume storage and S3 for backup/archival purposes. Our solution is flexible and future-proof, we want to be able to adapt to an ever-evolving technology and business landscape. Each step has been carefully considered based on the needs of the client, with the result being a more efficient, secure, and modern infrastructure.

Network Considerations

In order to aid the client with migrating their old infrastructure over to AWS, the network infrastructure must evolve to ensure scalability, performance, and cost-efficiency while maintaining reliable connectivity across all offices and optimizing cloud-based services. The company's 20 offices are currently connected to the corporate headquarters via site-to-site VPN. As we transition to AWS, we will assess the options available for better integration and reliability between on-premises systems and the cloud. Shifting over from the client's VPN, AWS offers its own VPN and Direct Connect. I believe Direct Connect would be the best fit. AWS Direct Connect provides a dedicated, high-bandwidth, low-latency connection to AWS which would be ideal for a company setting where we can maximize performance.

AWS' VPN, on the other hand, is an option but it's not as attractive. It is easier to set up and has lower initial costs, but in the long run, the cost can end up being higher for data transfer. AWS' VPN is also not as secure as direct connect, as direct connect is a private connection while the VPN relies on encryption. Direct connect is ideal for mission-critical applications requiring consistent network performance, such as ERP and database services. All this being said though, if cost is an issue, AWS VPN is still an extremely viable option instead of Direct Connect.

Assessing the current bandwidth and latency is essential to ensure that cloud applications meet the performance requirements. We may need to upgrade internet links or implement MPLS for higher performance if Direct Connect is not viable. To ensure high availability and resilience, we will implement redundant VPN tunnels and consider using AWS Transit Gateway for efficient routing and failover across multiple locations.

This will allow seamless communication between AWS resources and on-premises infrastructure which will minimize network downtime.

In order to optimize traffic and improve network performance, scalability, and reliability, we can integrate SD-WAN. SD-WAN will be able to provide intelligent routing between AWS and on premises resources. SD-WAN can help maintain optimal performance, even with fluctuating network conditions, ensuring critical traffic is prioritized.

As applications migrate to AWS, we will utilize Elastic Load Balancing (ELB) to distribute traffic across multiple instances of critical applications. ELB ensures that no single server becomes a bottleneck, improving the availability and fault tolerance of services. To prioritize critical applications such as ERP and database queries, we must implement AWS QoS policies. This will help ensure that high-priority traffic is processed first, even when network congestion occurs.

We can utilize Amazon Virtual Private Cloud (VPC) to create isolated subnets for different workloads to improve security and traffic management. This allows us to implement tighter security controls and enforce segmentation between various services. Security Groups and Network ACLs will be used in conjunction with AWS VPC to control traffic between subnets and external networks. To ensure data integrity and security, we will enforce encryption for all data transfers using IPsec VPN for on-premises connections and TLS for cloud services. AWS offers an elastic infrastructure that can scale with the business's growth.

We will also take advantage of Amazon EFS for scalable storage as the company itself continues to scale up. Amazon S3 can be used for backups and any desired archives, ensuring long-term retention and secure storage.

Finally, we can use Amazon Aurora for high-performance relational databases as the business grows and requires more database capacity. To prevent any single point of failure, we will need to deploy critical applications like ERP and email across multiple AWS Availability Zones for high availability and fault tolerance. I will once again point to Amazon S3 for backup where, in the case of failure, secure backups are present.

Client Considerations

With this migration, we must also consider the client's needs. The primary goal of the migration will be ensuring that client access to critical applications such as ERP, email, and internal web applications is secure, efficient, and reliable. All desktops and workstations will also need to meet the system requirements to interact with AWS-hosted applications, as this is what they'll be migrated over to.

This will include ensuring compatibility with cloud-based ERP systems and email solutions. To provide an optimal user experience, we will ensure all of the 100 desktops are updated to handle cloud-based workloads effectively. We will also implement AWS Systems Manager to automate software deployments, updates, and security patches. This will benefit the client because whether at remote offices or in the office space, we can ensure everyone's workstations are up to date. This solution will also ensure all devices are secure, minimizing the risk of any potential security vulnerabilities.

With the ERP and database hosted in AWS, secure and high-performance access must be ensured for all users. We will use AWS Direct Connect or AWS Site-to-Site VPN to create a stable and low-latency connection between on-premises systems and the cloud-hosted resources. This ensures reliable access to ERP and database services across all offices.

Amazon WorkMail will be used to manage email services across the organization. With WorkMail, email access will be highly available and scalable, providing seamless connectivity regardless of location. Email users will have access via webmail or mobile apps, ensuring accessibility across devices.

To optimize performance across all offices, we will deploy Amazon CloudFront (CDN) to accelerate the delivery of static content, such as internal websites. In terms of dynamic content, CDN is also able to handle this, however, we may want to consider AWS Global Accelerator. AWS Global Accelerator focuses on application performance and availability, so the choice between these two depends on the needs of the client. AWS Global Accelerator can be used to ensure low-latency access by routing traffic to the nearest regional endpoint.

To protect all endpoints accessing AWS resources, we will integrate AWS Security Hub with endpoint security solutions. This will allow for centralized visibility and management of device security across the organization.

We will also enforce data encryption for all data in transit and at rest using AWS Key Management Service (KMS). AWS provides extensive compliance tools, and we can leverage AWS Config to monitor and track the security of our infrastructure. We will implement stringent access controls and encryption protocols for sensitive departments (HR, Finance), ensuring the company meets regulatory and industry-specific requirements.

We will implement Amazon Connect to provide a centralized IT support system for all users, including troubleshooting tools, ticketing systems, and a knowledge base for resolving common issues. With all these new AWS systems in place, the employees will need to develop a familiarity with the applications, so proper training is also necessary.

After this, the client should see a massive improvement in all facets of their infrastructure overall.

Active Directory and DNS Servers

As part of a broader initiative to migrate enterprise infrastructure to AWS, the organization must modernize its foundational identity and network services like Active Directory (AD), Domain Name System (DNS), and Dynamic Host Configuration Protocol (DHCP). These services are currently hosted on four physical services across a centralized datacenter and 20 remote offices. Each server is configured with a single-socket-quad-core Xeon processor, 32GB of RAM, and 2x2TB drives in RAID 1 for redundancy (2TB usable). While the current setup has performed reliably, it lacks scalability, elasticity, and resiliency required for growth and cloud-native operations.

For this section I will be discussing a well targeted migration strategy and recommendations to transition to these core services to AWS, leveraging AWS Managed Microsoft AD, and Route 53 Private Hosted Zones, and VPC DHCP Option Sets. The proposed migration path aims to reduce dependency on aging infrastructure, improve operation availability, and enable integration with other cloud-hosted services such as ERP platforms, databases, and enterprise email systems.

The current AD infrastructure provides authentication, group, policy management, and user identity services across the enterprise. With four servers handling these duties in addition to DNS and DHCP, the system is burdened with multi-tiered roles that complicate scalability and recovery.

Recommendations for the migration plan leveraging the use of AWS Managed Microsoft AD (Enterprise Edition). This service supports up to 500,000 directory objects and is designed for highly available, multi-AZ deployments. According to AWS, Managed Microsoft AD supports forest trusts relationships and native compatibility with existing on premises AD domains, making it well suited for hybrid migrations **(What Is AWS Directory Service? - AWS Directory Service, n.d)**. This trust model allows for a hybrid deployment where authentication continues to work seamlessly across legacy systems and new AWS hosted workloads. Establishing trust relationships also eliminates the need to synchronize users or duplicate credentials, simplifying identity management while preserving security controls.

Currently, internal DNS is handled by the same AD Integrated servers. These systems provide name resolution for all networked clients, including office workstations, file shares, application servers, and internal domains. A recommendation to consider for migration is deploying Amazon Route 53 Private Hosted Zones to handle DNS resolution for AWS based services **(Amazon Web Services, 2023)**. Route 53 can be configured with conditional forwarders to route specific DNS queries back to the on premises environment, preserving resolution consistency during hybrid operations. This ensures that both on premises and AWS resources can resolve internal names without conflict, particularly during the transitional phase of the migration. Using Route 53 in place of manually-managed DNS zones also enables centralized control, logging, and scalability, while offloading the administrative burden of maintaining DNS infrastructure.

The organization currently delivers DHCP services through the same AD/DNS servers. These servers issue IP configurations to thousands of devices, including workstations, printers, and network appliances across multiple remote locations. A recommendation to consider for migration is introducing VPC DHCP Option Sets for all AWS-hosted resources, such as EC2 instances and internal cloud services. These options can be configured to specify DNS servers (such as AWS Managed Microsoft AD), domain names, and other necessary parameters **(DHCP Options Sets for Your VPC - Amazon Virtual Private Cloud, n.d)**. DHCP Option Sets will ensure consistency and centralized control over IP addressing in AWS. For remote offices and on premises devices, retain existing DHCP services temporarily to avoid service disruption. As offices transition to AWS based network architectures (via VPN or Direct Connect), DHCP responsibilities can be phased out and consolidated within AWS reducing reliance on physical infrastructure.

Justification for migrating current services to AWS

Current Service	AWS Solutions	Justification
AD Domain Services	AWS Managed Microsoft AD	Scalable, hybrid-ready authentication
DNS (Internal)	Route 53 Private Hosted Zones	Reliable name resolution with conditional forwarding
DHCP (Client IPs)	VCP DHCP Option sets + On-prem DHCP (temporary)	Gradual transition, maintains IP consistency

Email Servers

The organization's email infrastructure comes next after the DNS and Active Directory services have been successfully migrated. This guarantees that before moving important communication systems, user authentication and access policies are completely established. Four Windows Server 2008 R2 email servers make up the current configuration, which supports two distinct domains: one for upper management and one for non-management staff. The organization's datacenter infrastructure has 45TB of usable storage thanks to each server's quad socket quad-core Xeon processors, 96GB of RAM, and 16×3TB drives in RAID 5.

We advise switching to Amazon WorkMail, AWS's managed email and calendaring service, to update this system and improve scalability, dependability, and accessibility. WorkMail offers robust compatibility with Active Directory and facilitates smooth integration with mobile email clients and Microsoft Outlook. Users will still be able to access their emails on desktop or mobile platforms with WorkMail, which also offers enhanced redundancy and cloud-native features like maintenance and update automation.

With yearly growth rates of 25% and 30%, respectively, the company currently manages about 4000 non-management accounts (10GB each, 40% utilization) and 150 upper management accounts (each with 100GB quotas, 60% utilization). With its pay-as-you-go pricing, automatic scalability, and elimination of the need to provision new hardware or physical storage, Amazon WorkMail will make growth management easier.

If the company needs outbound email automation for services like billing, HR notifications, or customer engagement, AWS Simple Email Service (SES) can be integrated to manage high-volume transactional or marketing emails for more flexibility. Mission-critical communications benefit greatly from SES's dependable throughput.

Another big worry when migrating emails is security. Encryption of data in transit and at rest is ensured by integration with AWS Key Management Service (KMS). Fine-grained access controls can also be enforced by AWS Identity and Access Management (IAM). AWS CloudTrail can be used to activate audit logs for all user and administrative actions pertaining to email accounts and configurations in order to ensure compliance.

To ensure high availability and fault tolerance, email workloads will be distributed among multiple AWS Availability Zones. Disaster recovery is made easier and downtime is decreased with this architecture. Setting up backups with lifecycle policies that align with required retention and, if required, allowing archival storage through Amazon Glacier are both possible with Amazon S3.

By switching to AWS WorkMail and SES, the business will improve long-term performance, streamline operations, improve security and compliance, and get rid of outdated on premises infrastructure, opening the door for a cutting edge, cloud first communication platform.

ERP/Database Servers

The ERP and database servers will be moved to AWS in order to modernize business critical systems after email services have been migrated. Eight mid-frame servers make up the current on premises infrastructure: two are used for development and testing, and six are used for production. With quad-socket quad core Xeon processors, 96GB of random-access memory, and 8x2TB drives in RAID 5, each server has roughly 6TB of usable storage.

Inventory, accounting, human resources, and customer relationship management are just a few of the operational tasks that are managed by the company's Oracle-based ERP system, which is supported by these servers. However, the scalability and resilience needed to support peak workload demands and rapid business growth are lacking in this hardware setup, which is approaching obsolescence.

Depending on licensing and feature compatibility, we advise moving the ERP workloads to Amazon Relational Database Service (RDS) for Oracle or Amazon Aurora in order to support a cloud-first strategy. Both services offer scalable, managed database solutions with integrated high availability, monitoring, and backups. In addition to supporting read replicas, automated failover, and replication across multiple AWS Availability Zones for disaster recovery, Aurora provides improved performance up to five times faster than standard MySQL.

Amazon EC2 instances with Oracle manually installed may be utilized by businesses that would rather have direct control over the database server environment. While still utilizing cloud-native features like virtual networking, elastic storage, and snapshot backups, this method offers the flexibility of custom configurations.

AWS Database Migration Service (DMS) can be used to set up real-time data replication between on-premises Oracle servers and AWS hosted RDS or EC2 databases in order to preserve continuity throughout the migration process. This guarantees the least amount of disturbance and permits comprehensive testing prior to complete cutover.

Throughout this migration, security and compliance continue to be top priorities. AWS Key Management Service (KMS) will be used to manage encryption for both data in transit and at rest, and AWS Identity and Access Management (IAM) policies will be implemented to safeguard sensitive data access. AWS CloudTrail can also be used to keep an eye on activity logs in order to guarantee auditability.

The company will benefit from increased performance, enhanced scalability, automated backups, and cost-effectiveness through on-demand provisioning by migrating its database and ERP systems to AWS. This step improves the overall agility of business operations by future-proofing core systems and facilitating improved integration with other migrated services, such as email and application servers.

Application Servers

Currently, the application servers run across 50 dedicated servers, each supporting various internal services such as ERP connections, time reporting, HR requests, and IT support tools. These servers operate on a mix of middleware and data-retaining setups, all running on Windows Server 2008 R2. Given the age of the operating systems and hardware, this AWS migration presents a valuable opportunity to move these workloads to Amazon EC2 to improve performance, simplify management, and ensure continued support and security.

For this migration, we decided to take a lift-and-shift approach using EC2, which allows us to replicate the existing server environment in the cloud with minimal changes. This approach is ideal for the current application stack since many of the services are tightly integrated with Windows and legacy components that are not easily containerized without a major refactor. EC2 gives us the flexibility to move fast while maintaining full control over the configuration and environment important for internal applications that have specific requirements.

By using EC2, we can modernize the infrastructure without needing to re-engineer the applications. It also gives us the ability to scale horizontally, meaning we can spin up additional servers as needed to meet increased demand, especially for middleware servers that handle high volumes of internal traffic. For services that rely on persistent storage, we can use Amazon EBS volumes for application data, keeping storage local to the server while still benefiting from AWS-managed durability and snapshot capabilities.

In terms of environment separation, we can mirror our current structure by creating a staging/test environment in a separate AWS account or VPC. This lets us validate updates and changes in isolation before pushing them into production. Testing environments can be spun up or down on demand to save cost, which is a significant improvement over maintaining physical test servers.

For compute and memory requirements, each application server will be provisioned using t3.large EC2 instances, which provide 2 vCPUs and 8 GB of RAM matching the current on-prem setup. This ensures consistent performance while allowing room to scale vertically or horizontally as application needs evolve. For middleware servers, we can also explore using auto-scaling groups to dynamically adjust based on load, bringing in some of the cloud-native benefits without needing to completely redesign the applications.

Overall, migrating the application servers to EC2 gives us a reliable, cost-effective path forward while maintaining the flexibility to modernize further over time.

Web Server Migration

Currently, the web server runs on 5 different servers, three IIS servers and two on Linux. These servers host both internal and external-facing web applications that serve different functions. For this AWS migration, a solid option is to use Amazon EC2 instances to host these servers while maintaining compatibility with the existing applications. This migration approach allows us to preserve functionality while still gaining benefits such as increased flexibility, scalability, and modernization over the older on-premises setup.

For this effort, we decided to replicate the current setup using EC2, making it a lift-and-shift operation. While services like Fargate, ECS, and EKS offer more modern approaches, they also require additional steps like containerizing applications and reconfiguring workloads. Given the current infrastructure and application stack particularly the use of IIS and older configurations, we determined that EC2 provides the best balance between modernization and ease of migration. It allows us to get up and running quickly while giving us the flexibility to modernize further in the future if needed.

Additionally, EC2 allows us to maintain direct control of the OS, networking, and configuration, which is important since many of our existing web applications are tightly coupled to the server environment they were built on. We can also take advantage of Amazon’s features like Elastic Load Balancing, auto-scaling, and EBS volume snapshots to bring in more modern operational capabilities without overhauling the existing application stack.

Beyond just hosting the web servers, we also need to consider other factors like storage, environment separation, and resource requirements. To support testing and development efforts, we can use a separate AWS account or isolated VPCs to run non-production instances. This allows us to safely test updates and features before promoting them to production.

In terms of storage, Amazon EBS will be used to provide persistent block-level storage to each EC2 instance. EBS is ideal for this type of lift-and-shift environment, it provides reliable performance and integrates well with EC2. We can configure EBS volumes for both the operating system and the application data separately, similar to the current setup. Finally, to meet compute and memory requirements, we can use t3.large instances which provide 2 vCPUs and 8 GB of RAM matching our current environment while giving us flexibility to scale up if needed in the future.

Storage Server Specifications

The current infrastructure's initial analysis mentions that, within the system, there are 8 GB of RAM present per server. Additionally, there are 8x2 TB drives, one dedicated to operations concerning the operating system and one dedicated to application data and application storage. With 15 servers being present in the infrastructure, there would be a total of 120 GB of RAM required for the system as well as 240 TBs in the servers' collective drives. Combining these results, a total of 240.12 TB of storage would be needed for the current infrastructure as it migrates from its current form to the cloud.

Each server is also fitted with a single-socket, dual core Xeon-class processor. Each processor is designed to work with a singular motherboard (contained within each server) whilst holding two cores on each CPU, which allows for multiprocessing and parallel programming (which contributes to a more efficient scheduling/task-execution process). Xeon-class processors are designed specifically for server operations, however, Xeon-class processors have several disadvantages with their usage. Xeon processors have higher cooling requirements, which directly contributes to a notably higher power consumption when they are run. Their CPUs are also slower as a result of the multiple cores present within them, so their clock speeds underperform compared to similar processors.

Transferring the current infrastructure from these Xeon processors to an AWS-based approach would greatly benefit the company because, assumedly, the processors are much more expensive (due to their heightened power consumption and cooling costs) than the AWS services we plan to use. Transferring to AWS over these physical servers would be both less expensive and more ergonomic for the company's advancing infrastructure.

In order to ensure we follow regulatory compliance during this migration process, we must backup the data in our system nightly. This can be done through the AWS Backup service (present in AWS S3), which can back up data on demand (via manual execution/commands) or automatically via a scheduled backup plan. We would easily follow the required scheduling parameters using the scheduled backup plan, detailing the plan to backup at a certain time every night to ensure nightly backups are carried out whilst retaining the backup data for the aforementioned 30 day period.

Moreover, AWS S3 provides a service known as “Cross-Region Replication” (CRR) that assists in data replication between distant locations using this service, which greatly improves regulatory compliance in the case of infrastructures needing to store data at notable distances, which would greatly improve this migration if these departments are set up in an interstate or international fashion.

Conclusion

Through our research of various cloud services available, we have found that Amazon Web Services would provide the most efficient, convenient, and personalized services for our current infrastructure over the course of (and following) the migration. Several functions provided by AWS, such as AWS S3, Amazon CloudFront, and SD-WAN directly parallel the needs of our current infrastructure, making it unquestionably the most fitting choice moving forward. In conclusion, the support provided by the many applications of AWS would not only uphold the current operations carried out by the company, but would demonstrably streamline, modernize, and improve the workflow of these operations, allowing for a more flexible infrastructure as technology continues to advance.

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